

Unit 1:

Rectangular Coordinate System

What We'll Cover:

- 1. Get to Know Rectangular Coordinate Systems:** We'll explore what 2D and 3D coordinate systems are, why they're important, and how they work.
- 2. Working with Position Vectors:** Learn how to write position vectors in two ways - using their components (like $\langle x, y, z \rangle$) and as column matrices.

Introduction: The Need for a Coordinate System

Physical quantities, such as electric fields in electromagnetics or fluid velocities in fluid mechanics, often vary with both space and time. To study and understand how these quantities change across space, we need a way to uniquely identify every point within that space.

This is where coordinate systems play a crucial role. They provide a structured and consistent framework for defining points and describing spatial variations, making it possible to analyze these physical phenomena effectively.

What is a Coordinate System?

In geometry, a coordinate system provides a framework of reference lines or curves to uniquely define the position of points or geometric elements in 2D or 3D space.

In this presentation, you will learn how to:

- Represent positions in the 2D and 3D Cartesian (or Rectangular) Coordinate Systems as position vectors.
- Calculate the distance between two points within the coordinate system.
- Add, subtract, scale, and multiply physical vectors.

The Cartesian Coordinate System

The Cartesian Coordinate System is one of the simplest and most widely used systems for defining positions in space. It consists of straight, graduated lines called **coordinate axes** that are oriented perpendicular to each other. Depending on the application, this system can operate in either 2-dimensional (2D) or 3-dimensional (3D) space to define points.

2-D Cartesian Coordinate System

In a 2-D Cartesian coordinate system, a point A is described by two real numbers, called *coordinates*. These numbers represent the distances of the point's perpendicular projections onto two fixed, perpendicular axes - commonly labeled as the x -axis and y -axis - which intersect at the origin $(0,0)$ (see Figure 1).

For example, if a point A is located at a distance a_x along the x -axis and a_y along the y -axis, it is assigned the coordinates (a_x, a_y) .

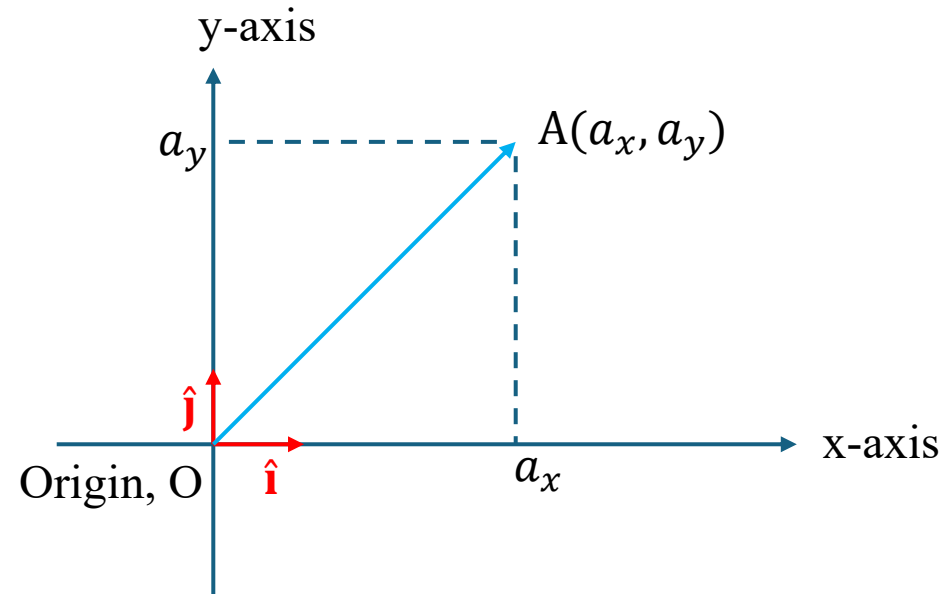


Figure 1: A point $A(a_x, a_y)$ in the 2-D Cartesian Coordinate System.

The position vector of a point A, with coordinates (a_x, a_y) , is defined as the directed distance from the origin O to A. This vector is denoted as $\overrightarrow{OA}$, where the arrow above the letters signifies that it represents a directed line segment, specifically pointing from O to A. When expressed in its standard unit vector form, the 2-D position vector $\overrightarrow{OA}$ is written as:

$$\overrightarrow{OA} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}}$$

Here, $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are the unit vectors pointing in the x- and y-directions, respectively.

In **column vector form**, position vector $\overrightarrow{OA}$ is represented as:

$$\overrightarrow{OA} = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$$

Note:

We typically express position vectors in **column vector** form because column vectors are more convenient for vector operations like dot products, cross products, and matrix transformations.

The **distance** from O to A is given by the magnitude of the position vector $\overrightarrow{OA}$:

$$d = |\overrightarrow{OA}| = \sqrt{(a_x)^2 + (a_y)^2}$$

The **direction** of position vector $\overrightarrow{OA}$ is described by the unit vector $\widehat{OA}$, which is defined as:

$$\widehat{OA} = \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|} = \frac{a_x\hat{\mathbf{i}}+a_y\hat{\mathbf{j}}}{\sqrt{(a_x)^2+(a_y)^2}}$$

Worked Example:

Given two points $A(3, 4)$, $B(7, 1)$, find the position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$.

Solution:

A plot the two points in the 2-D Cartesian coordinate system is shown in Figure 2.

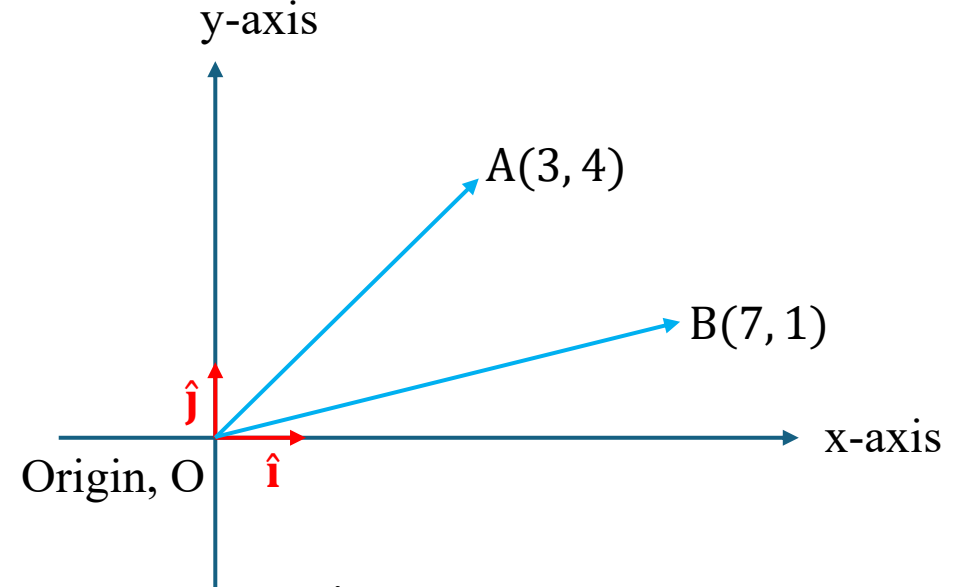


Figure 2

Directions of $\overrightarrow{OA}$ and $\overrightarrow{OB}$

To express the **direction of the position vectors** in terms of unit vectors, we first calculate the **unit vectors** in the direction of $\overrightarrow{OA}$ and $\overrightarrow{OB}$. A unit vector is a vector with a magnitude of 1, pointing in the same direction as the original vector.

1. Formula for a Unit Vector:

The unit vector $\widehat{OA}$ (say) corresponding to a position vector $\overrightarrow{OA}$ is given by:

$$\widehat{OA} = \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|},$$

where $|\overrightarrow{OA}|$ is the magnitude of $\overrightarrow{OA}$.

2. Magnitudes of the Vectors:

Write the position vectors of the points A and B as column matrices:

$$\overrightarrow{OA} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \overrightarrow{OB} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

- Magnitude of $\overrightarrow{OA}$:

$$|\overrightarrow{OA}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

- Magnitude of $\overrightarrow{OB}$:

$$|\overrightarrow{OB}| = \sqrt{7^2 + 1^2} = \sqrt{49 + 1} = 5\sqrt{2}$$

3. Unit Vectors:

- Unit vector in the direction of $\overrightarrow{OA}$:

$$\widehat{OA} = \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|} = \frac{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}{5} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$$

- Unit vector in the direction of $\overrightarrow{OB}$:

$$\widehat{OB} = \frac{\overrightarrow{OB}}{|\overrightarrow{OB}|} = \frac{\begin{bmatrix} 7 \\ 1 \end{bmatrix}}{5\sqrt{2}} = \begin{bmatrix} \frac{7}{5\sqrt{2}} \\ \frac{1}{5\sqrt{2}} \end{bmatrix}$$

4. Final Result:

- Direction of $\overrightarrow{OA}$ in terms of unit vectors:

$$\widehat{OA} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j} = 0.6\hat{i} + 0.8\hat{j} \blacksquare$$

- Direction of $\overrightarrow{OB}$ in terms of unit vectors:

$$\widehat{OB} = \frac{7}{5\sqrt{2}}\hat{i} + \frac{1}{5\sqrt{2}}\hat{j} = 0.99\hat{i} + 0.14\hat{j} \blacksquare$$

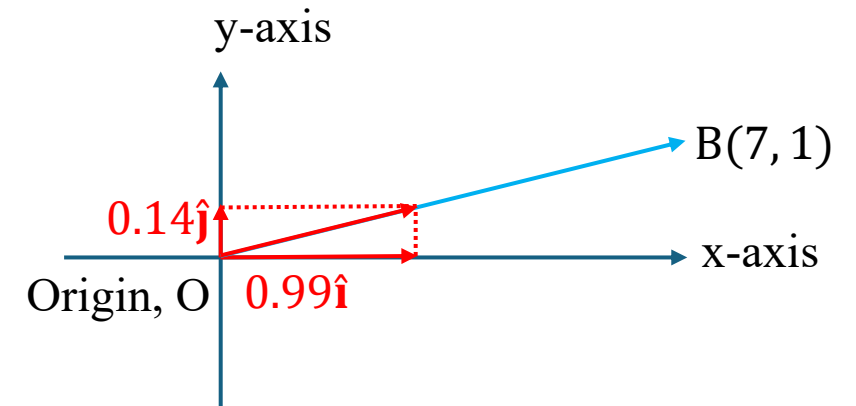
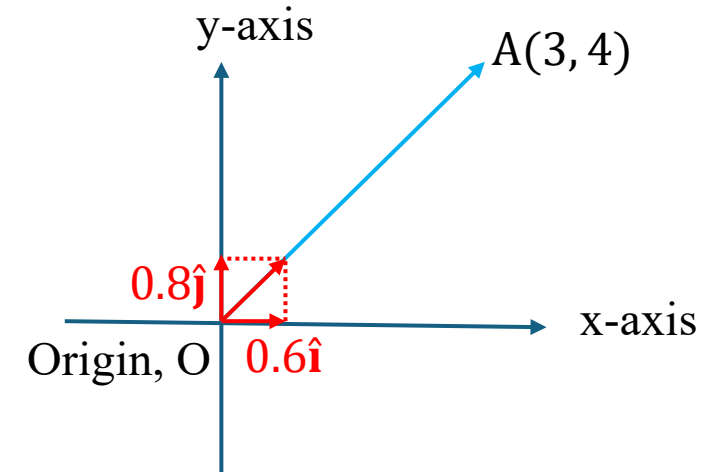


Figure 3

What is the 3D Cartesian Coordinate System?

The 3D Cartesian Coordinate System consists of three fixed, perpendicular, graduated axes that intersect at a common point called the **origin**. In a **right-handed** coordinate system, the axes follow the **right-hand rule** (refer to Figure 4):

- The positive x-axis points out of the page.
- The positive y-axis points to the right.
- The positive z-axis points vertically upward.

This setup provides a structured framework for defining the position of points in three-dimensional space.

Note: This course will only use the right-hand rule in labelling the axes.

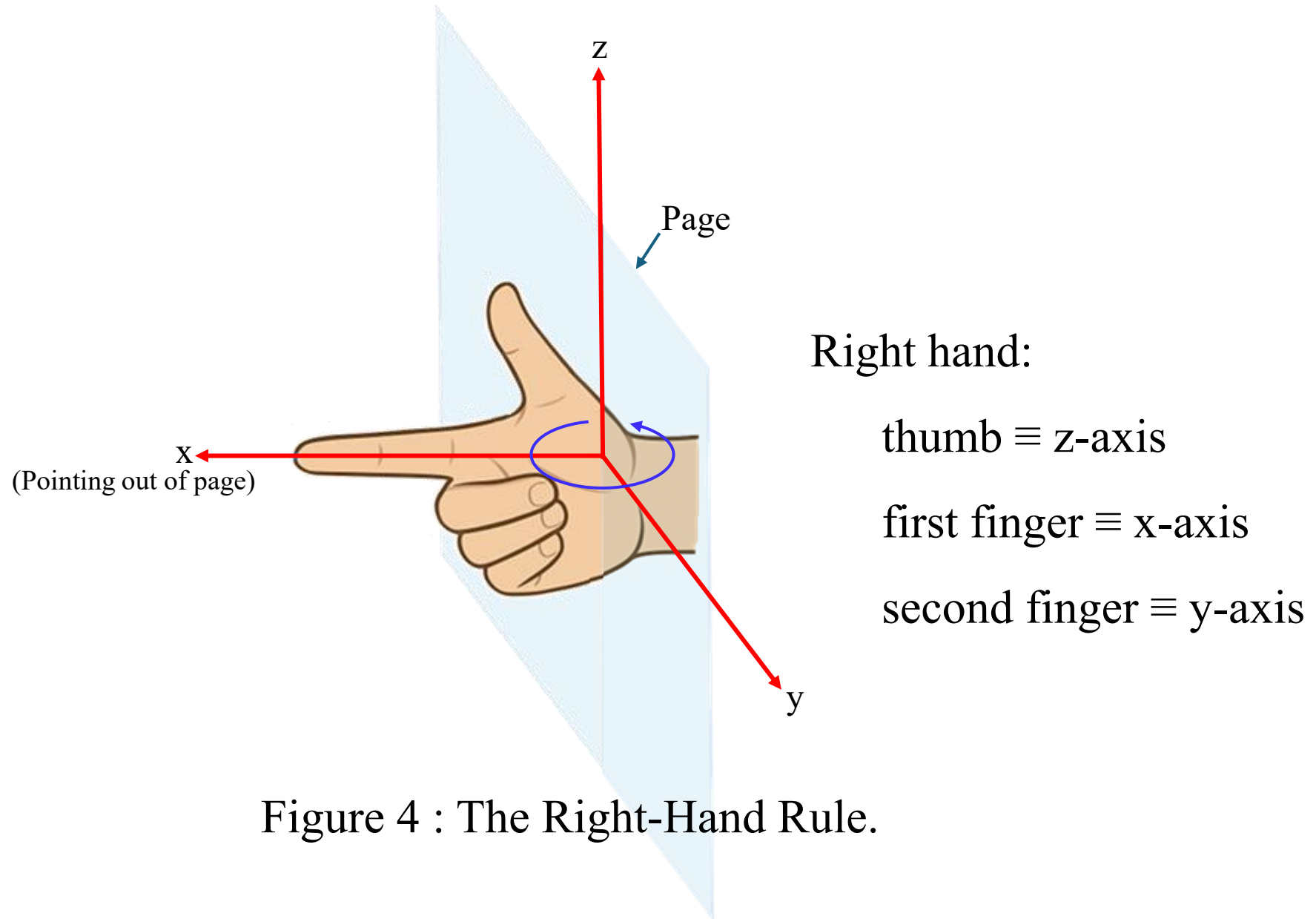


Figure 4 : The Right-Hand Rule.

COORDINATE PLANES

The planes associated with each pair of axes are shown in Figure 5.

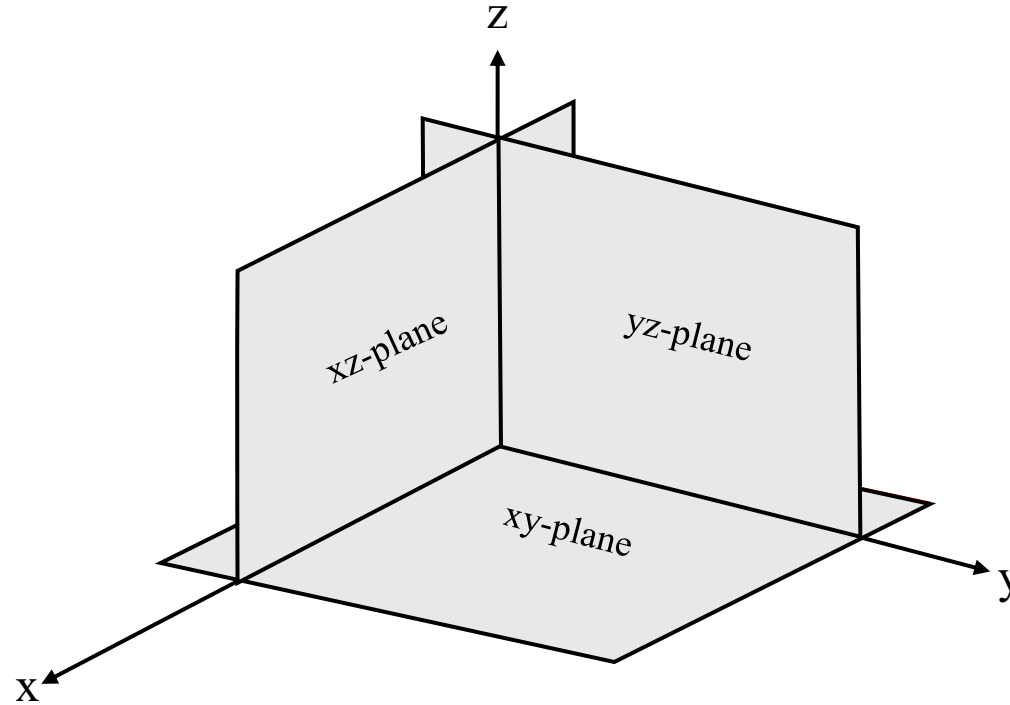


Figure 5: Coordinate planes

A point P in a 3-D Cartesian coordinate system is represented by a 3-tuple of real numbers (a, b, c) , which specify the positions of the perpendicular projections of the point onto the three axes. In Figure 6 below, a denotes the x-coordinate of point P , b denotes the y-coordinate, and c denotes the z-coordinate.

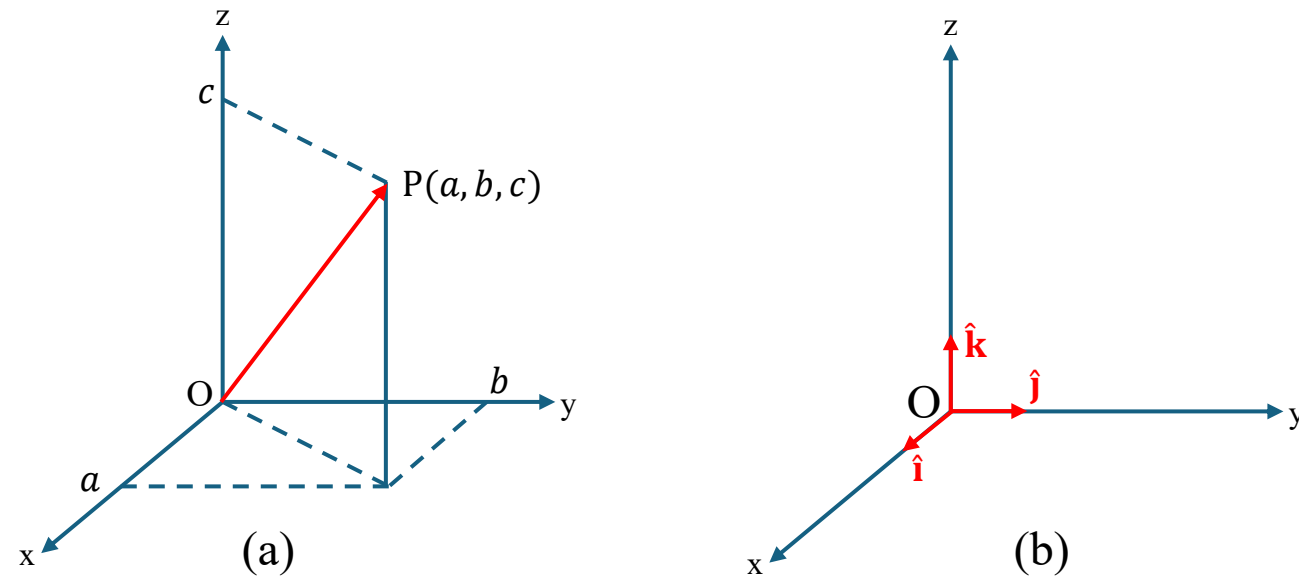


Figure 6: 3-D Cartesian Coordinate System (a) A point in the 3D coordinate system, (b) Unit vectors in the 3D coordinate system.

The 3-D position vector $\overrightarrow{OP}$, when expressed in standard unit vector form, is written as:

$$\overrightarrow{OP} = a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}}$$

where $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are unit vectors pointing in the positive x-, y-, and z-directions, respectively. In column vector form, it is written as:

$$\overrightarrow{OP} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Similar to the 2-D case, the distance from O to P in the 3-D case is given by the magnitude of the vector $\overrightarrow{OP}$. That is, the distance from O to P is:

$$d = \sqrt{a^2 + b^2 + c^2}$$

Likewise, the direction of the vector $\overrightarrow{OP}$ is given by the unit directional vector $\widehat{OP}$, defined as follows:

$$\widehat{OP} = \frac{\overrightarrow{OP}}{|\overrightarrow{OP}|} = \frac{a\hat{i}+b\hat{j}+c\hat{k}}{\sqrt{a^2+b^2+c^2}}$$

Notes

Some literature adopts an alternative notation for the rectangular unit vectors, using $\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_y$, $\hat{\mathbf{a}}_z$ in place of the more common $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ symbols. Thus, instead of writing a vector $\mathbf{A}$ as

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}},$$

it may alternatively be expressed as

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z.$$

Both notations are equivalent and represent the same vector components along the $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ axes, respectively. The choice of notation typically depends on the convention used in a particular textbook or discipline.

Worked Example 1:

Given a point $P(-1, 2, 3)$ in the 3-D Cartesian coordinate system (see Figure 7), express the position vector $\overrightarrow{OP}$ in both standard unit vector form and column vector form.

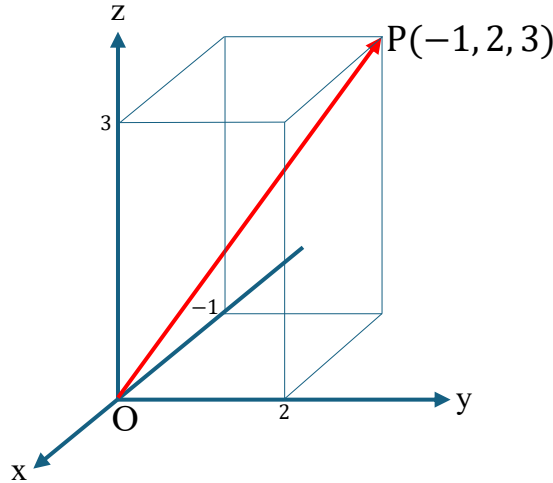


Figure 7: 3-D Cartesian Coordinate System

Solution

1. Standard Unit Vector Form: The position vector $\overrightarrow{OP}$ is:

$$\overrightarrow{OP} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

Here $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the unit vectors pointing in the positive x-, y-, and z-directions, respectively.

2. Column Vector Form: The position vector $\overrightarrow{OP}$ in column vector form is:

$$\overrightarrow{OP} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$$

Physical Vector

Up to this point, we've explored **position vectors**. By defining them as a type of physical vector, we recognized that their magnitude represents the distance from the origin to the endpoint, while their direction indicates the orientation of the directed line segment in a coordinate system.

We now aim to expand our understanding of physical vectors and discuss their similarities and differences compared to position vectors.

Examples of Physical Vectors

1. **Force:** A vector that represents a push or pull acting on an object. The magnitude of the force indicates its strength, and its direction specifies the line along which the force is applied (see Figure 8).

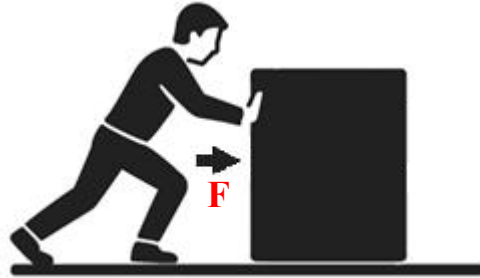


Figure 8: Push force as a vector $\mathbf{F}$ acting along a line

2. **Velocity:** A vector that represents the rate of change of an object's position. The magnitude of the velocity indicates the speed of the object, while its direction specifies the direction of motion (see Figure 9).

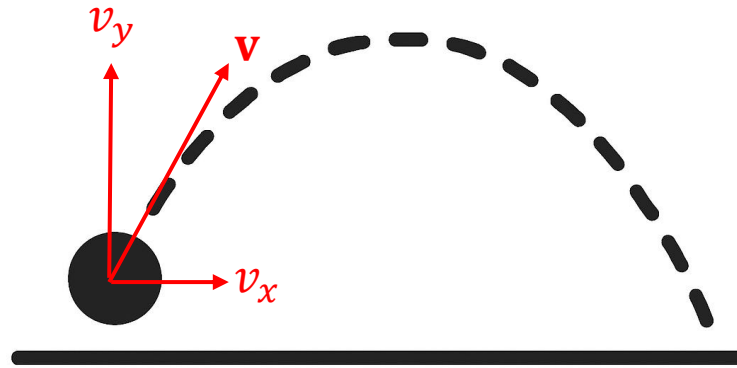


Figure 9: Object velocity $\mathbf{v}$ as a vector.

3. **Acceleration:** A vector that represents the rate of change of velocity with respect to time. The magnitude of acceleration indicates the rate of change of speed, while its direction specifies the direction of the change in velocity, which is often aligned with the net force causing the acceleration (see Figure 10).

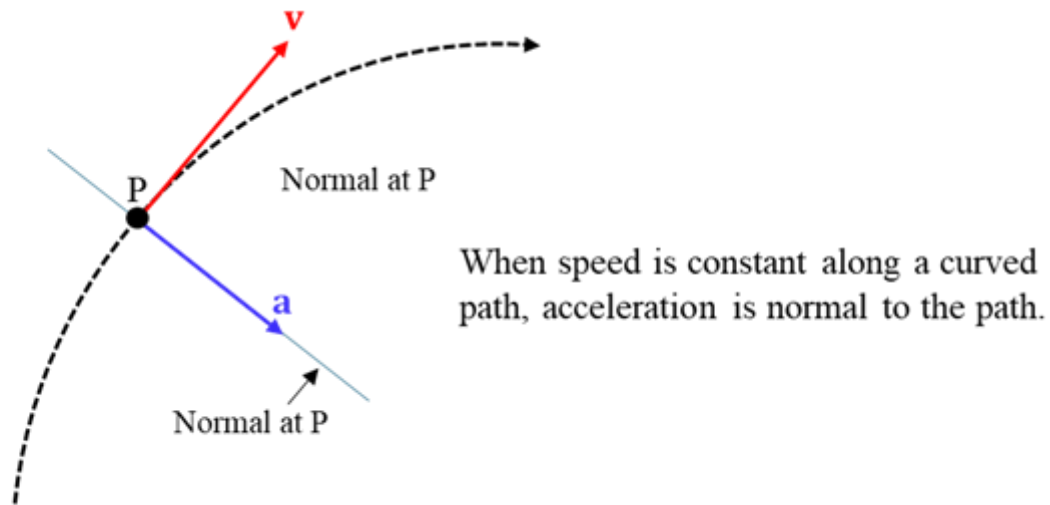


Figure 10. A particle moving along a curved path at constant speed v . Acceleration vector a is normal to the path.

4. **Momentum:** A vector equal to the product of an object's mass and its velocity, describing the motion of the object (see Figure 11).

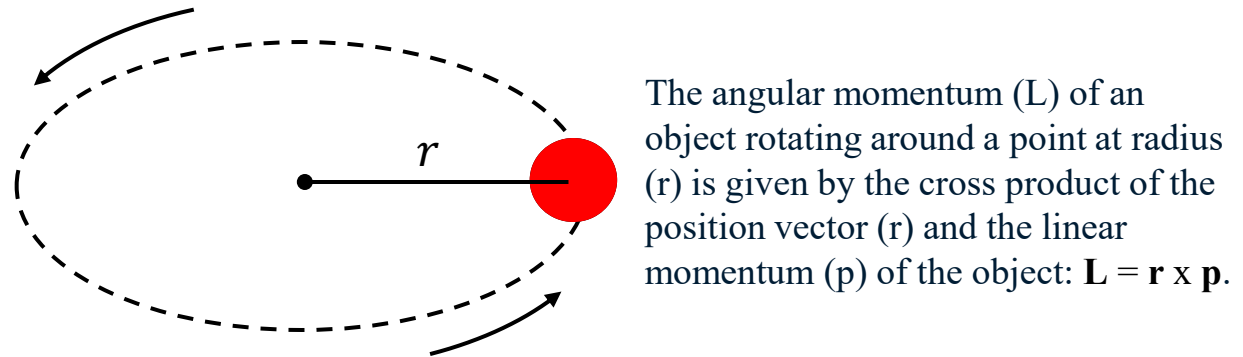
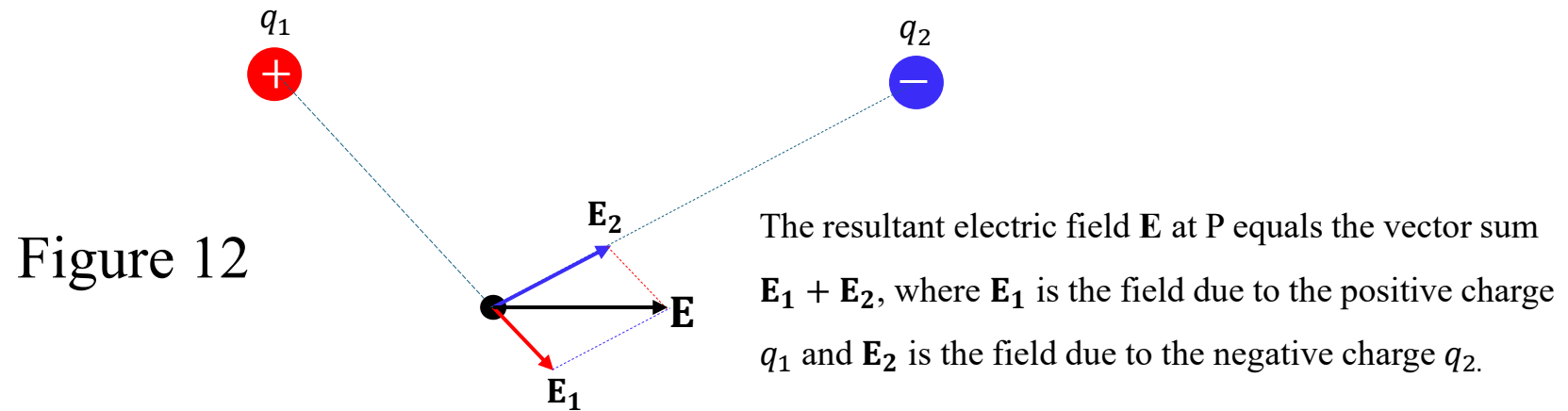
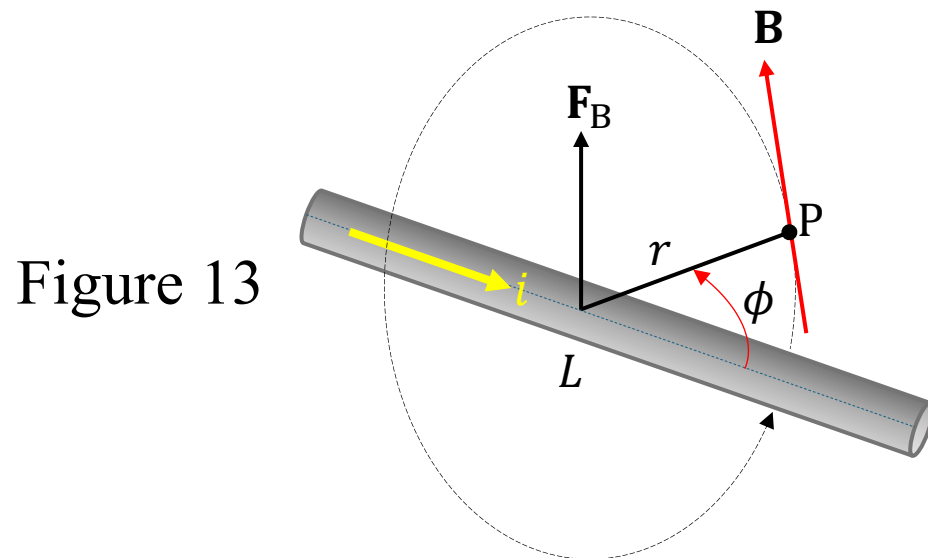


Figure 11: Object rotating around a point at radius r .

4. **Electric Field $\mathbf{E}$:** A vector field that describes the electric influence on both static and moving electric charges. The magnitude of the electric field represents the force experienced by a positive unit charge placed at a point in the field, while its direction indicates the orientation of the force acting on the charge (see Figure 12).



5. **Magnetic Field $\mathbf{B}$:** A vector field that describes the magnetic influence on moving electric charges and magnetic materials. The magnitude of the magnetic field at a point represents the strength of the field at that location, while its direction is defined as the direction in which the north pole of a compass needle would point if placed in the field (see Figure 13).



At a point P , located a distance r from the axis of a straight current-carrying conductor, the magnetic field vector $\mathbf{B}$ is directed along the tangent to the circular magnetic field line at P . The direction of the field is given by the right-hand rule: if the thumb points along the current, the curled fingers indicate the orientation of $\mathbf{B}$.

Similarities

1. Magnitude and Direction:

Both physical vectors and position vectors have a magnitude (a scalar quantity) and a direction in space.

2. Representation:

Both can be represented geometrically as arrows, where the length corresponds to the magnitude and the arrowhead indicates direction.

Similarities (continued)

3. Mathematical Description:

Both are described within a coordinate system (e.g., Cartesian, polar). For example, in 3D Cartesian coordinates, they are expressed as:

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

4. Operations:

They both support standard vector operations like addition, subtraction, dot product, and cross product (See Unit 2).

Differences

1. Purpose and Interpretation:

- **Position Vectors:**

A position vector defines a specific point in space relative to an origin. For

example, $\overrightarrow{OP} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ represents the position of a point P at (2, 2, 1). It is primarily used to describe locations in a coordinate system.

Differences

1. Purpose and Interpretation: (continued)

- **Physical Vectors:**

Physical vectors represent quantities like force, velocity, acceleration, or electric field, which have a magnitude and direction but are not tied to a specific point of origin. They describe changes or effects in a system.

Differences (continued)

2. Dependence on Origin:

- A position vector is always defined with respect to a fixed origin in a coordinate system.
- A physical vector does not require a specific origin unless its point of application or line of action is critical (e.g., torque).

Differences (continued)

3. Frame of Reference:

- A position vector changes if the origin shifts or if the coordinate system rotates or translates.
- Physical vectors may remain invariant in magnitude and direction under translations, though they can transform under rotations depending on the context.

Differences (continued)

4. Nature of the Quantity:

- Position vectors inherently describe a point or location in space.
- Physical vectors typically describe a dynamic or physical phenomenon, such as the rate of change of position (velocity) or a force applied to an object.

Understanding these similarities and differences is crucial when transitioning from physical intuition to mathematical modeling of vectors in fields like physics, engineering, and mathematics.

END OF UNIT 1